INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

BCS-501	FORMAL LANGUAGE AND AUTOMATA THEORY	L T P 3 0 0	3 Credits

Course Objectives:

CO1	To learn fundamentals of Regular and Context Free Grammars and Languages
CO2	To understand the relation between Regular Language and Finite Automata and machines.
CO3	To learn how to design Automata's and machines as Acceptors, Verifiers and Translators.
CO4	To understand the relation between Contexts free Languages, PDA and TM.
CO5	To learn how to design PDA as acceptor and TM as Calculators.

Detailed Syllabus

MODULE-I

Finite Automata and Regular Expressions: Finite State Systems, Basic Definitions NonDeterministic finite automata (NDFA), Deterministic finite automata (DFA), Equivalence of DFA and NDFA Finite automata with E-moves, Regular Expressions, Equivalence of finite automata and Regular Expressions, Regular expression conversion and vice versa, Arden's Theorem. Introduction to Machines: Concept of basic Machine, Properties and limitations of FSM.Moore and mealy Machines, Equivalence of Moore and Mealy machines.

MODULE-II

Properties of Regular Sets: The Pumping Lemma for Regular Sets, Applications of the pumping lemma, Closure properties of regular sets, Myhill-Nerode Theorem and minimization of finite Automata, Minimization Algorithm. Grammars: Definition, Context free and Context sensitive grammar, Ambiguity regular grammar, Reduced forms, Removal of useless Symbols and unit production, Chomsky Normal Form (CNF), Griebach Normal Form (GNF).

MODULE-III

Pushdown Automata: Introduction to Pushdown Machines, Acceptance of PDA, PDA to CFG and CFG to PDA, Application of Pushdown Machines Turing Machines: Deterministic and Non-Deterministic Turing Machines, Design of T.M,Halting problem of T.M., PCP Problem. Chomsky Hierarchies: Chomsky hierarchies of grammars, unrestricted grammars, Context sensitive languages, Relation between languages of classes. Computability: Basic concepts, Primitive Recursive Functions.

Text Book:

1. Introduction to automata theory, language & computations- Hopcroaft&O.D.Ullman, R Mothwani, 2001, Addison Wesley

Reference Books:

1. Theory of Computer Science (Automata, Languages and computation): K.L.P.Mishra & N.Chandrasekaran, 2000, PHI.
2. Introduction to formal Languages & Automata-Peter Linz, 2001, Narosa.

CO1	Students will be able to define the mathematical principles behind theoretical computer science.
CO2	Students will be able to identify the different computational problems and their associated complexity.
CO3	Students will be able to differentiate and give examples for the different types of automata like finite automata, push down automata, linear bounded automata and Turing machine.
CO4	To apply the techniques of designing grammars and recognizers for several programming languages.
CO5	Students will be able to correlate the different types of automata to real world applications.